

Exponential Map

1.1 One-Parameter Subgroups and the Exponential Map

1.1.1 One-Parameter Subgroups

Definition 1.1.1

A **one-parameter subgroup of G** is defined to be a Lie group homomorphism $\gamma : \mathbb{R} \rightarrow G$, with $\mathbb{R}$ considered as a Lie group under addition.

Theorem 1.1.2 ► Characterization of One-Parameter Subgroups

Let G be a Lie group. The one-parameter subgroups of G are precisely the maximal integral curves of left-invariant vector fields starting at the identity.

Remark.

Given $X \in \text{Lie}(G)$, the one-parameter subgroup determined by X in this way is called the **one-parameter subgroup generated by X** . Because left-invariant vector fields are uniquely determined by their values at the identity, it follows that each one-parameter subgroup is uniquely determined by its initial velocity in $T_e G$, and thus there are one-to-one correspondences

$$\{\text{one-parameter subgroups of } G\} \longleftrightarrow \text{Lie}(G) \longleftrightarrow T_e G.$$

1.1.2 The Exponential Map

Definition 1.1.3 ► Exponential map

Given a Lie group G with Lie algebra $\mathfrak{g}$, we define a map $\exp : \mathfrak{g} \rightarrow G$, called the **exponential map of G** , as follows: for any $X \in \mathfrak{g}$, we set

$$\exp X = \gamma(1),$$

where γ is the one-parameter subgroup generated by X , or equivalently the integral curve of X starting at the identity .

Proposition 1.1.4 ►

Let G be a Lie group. For any $X \in \text{Lie}(G)$, $\gamma(s) = \exp sX$ is the one-parameter subgroup of G generated by X .

Proof. Let $\gamma : \mathbb{R} \rightarrow G$ be the one-parameter subgroup generated by X , which is the integral curve of X starting at e . For any fixed $s \in \mathbb{R}$, the rescaling lemma of integral curve gives that $\tilde{\gamma}(t) = \gamma(st)$ is the integral curve of sX starting at e , so

$$\exp sX = \tilde{\gamma}(1) = \gamma(s)$$

□

1.2 Normal Subgroups

Lemma 1.2.1

Let G be a connected Lie group, and let $H \subseteq G$ be a connected Lie subgroup. Let $\mathfrak{g}$ and $\mathfrak{h}$ denote the Lie algebras of G and H , respectively. Then H is normal in G if and only if

$$(\exp X)(\exp Y)(\exp(-X)) \in H \quad \text{for all } X \in \mathfrak{g} \text{ and } Y \in \mathfrak{h}. \quad (1.1)$$

1.2.1 The Adjoint Representation

Definition 1.2.2

Let G be a Lie group and $\mathfrak{g}$ be its Lie algebra. For any $g \in G$, the conjugation map $C_g : G \rightarrow G$ given by $C_g(h) = ghg^{-1}$ is a Lie group homomorphism. We let $\text{Ad}(g) = (C_g)_* : \mathfrak{g} \rightarrow \mathfrak{g}$ denote its induced Lie algebra homomorphism.

Proposition 1.2.3 ► The Adjoint Representation

If G is a Lie group with Lie algebra $\mathfrak{g}$, the map $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a Lie group representation, called the *adjoint representation* of G .

Definition 1.2.4

Given a finite dimensional Lie algebra $\mathfrak{g}$, for each $X \in \mathfrak{g}$, define a map $\text{ad}(X) : \mathfrak{g} \rightarrow \mathfrak{g}$ by $\text{ad}(X)Y = [X, Y]$.

Proposition 1.2.5 ▶ adjoint representation of $\mathfrak{g}$

For any Lie algebra $\mathfrak{g}$, the map $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is a Lie algebra representation, called the *adjoint representation of $\mathfrak{g}$* .

Theorem 1.2.6

Let G be a Lie group, let $\mathfrak{g}$ be its Lie algebra, and let $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ be the adjoint representation of G . The induced Lie algebra representation $\text{Ad}_* : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is given by $\text{Ad}_* = \text{ad}$.